Detecting AI Coding Agent Adoption

Every way to detect it, and every trace for every agent. Based on Robbes et al., *Agentic Much?* (2026), plus our own checks of GitHub on 24–25 October 2026.

> **How to read this.** Part 1 lists the ways to detect agents (where to look). Part 2 lists each agent with its exact traces. Text in `this font` is the exact string to match. Numbers in brackets are rough GitHub search counts (Oct 2026) to show how common a trace is. **No trace never means human**: every trace can be turned off, so any count is a minimum.

Part 1 — All the ways to detect an agent

1. Commit trailers · strong · this commit

- Lines at the end of the commit message. Read them with `git log --format=%B` or `git log --format=%(trailers)`.
- `Co-authored-by: <agent> <email>` is the most common. Match the **email**, not the name.
- Generic AI trailers: `Assisted-by: AGENT:MODEL` (Linux kernel rule), `AI-Assisted-By:`, `Generated-by:`.
- Tool-specific trailers: `Agent-Logs-Url:` (Copilot), `Claude-Session:` (Claude Code on the web), `Amp-Thread-ID:` (Amp), `Replit-Commit-Author: Agent` (Replit), `X-Lovable-Edit-ID:` (Lovable), `Made-with: Cursor`.

2. Commit author or committer · strong · this commit

- The agent is the author, e.g. `Cursor Agent <cursoragent@cursor.com>` or a bot such as `google-labs-jules[bot]`.
- Aider instead adds `(aider)` to your own name: `Jane Doe (aider)`.
- Bot emails look like `<id>+<login>@users.noreply.github.com`. The **id number** is hard to fake, so check it.

3. Text the agent writes in the message · strong · this commit

- Footers like `(robot emoji) Generated with [Claude Code](...)` and links to the agent session or task (`chatgpt.com/codex/tasks/`, `app.devin.ai/sessions/`, `ampcode.com/threads/`).
- Only count the exact footer or link. Brand words anywhere in a message are **not** a signal.

4. Pull requests · strong · this PR

- PR **author** is an agent bot (Copilot, cursor[bot], devin-ai-integration[bot], ...).
- PR **branch** starts with an agent prefix: `codex/`, `copilot/`, `cursor/`, `claude/`, `devin/` ...
- PR **label**: `codex`, `amp`, `open-swe`, `ai-generated`.
- PR **body** contains the agent footer or session link (Claude Code: ~19.5M PR bodies).
- When a PR is squash-merged its branch name is lost: copy the PR's label to the squash commit.

5. Issues and comments · medium · the repo uses the agent

- Issues **assigned** to an agent (Copilot, Jules, Devin, OpenHands). The issue is the agent's task.
- **@mentions** that start an agent in a comment: `@claude`, `@codex`, `@copilot`, `@cursor`, `@jules`, `@devin`, `@gemini-cli`.
- Comments written by agent bots (e.g. `claude[bot]` commented on ~101K issues).
- Careful: a mention can just be people talking about the tool. Prefer comments the bot answered.

6. AI code review · separate measure · review adoption

- Reviews or comments by AI reviewers: `copilot-pull-request-reviewer[bot]` (~6.3M PRs), `coderabbitai[bot]` (~8.4M), `gemini-code-assist[bot]` (~1.7M), `cursor[bot]` (Bugbot), `chatgpt-codex-connector[bot]`, `qodo-code-review[bot]`, `ellipsis-dev[bot]`, `greptile-apps[bot]`, `sourcery-ai[bot]`.
- They review code, they don't write it: report this as a separate kind of adoption.
- But a commit that applies their fix gets a co-author line (e.g. `coderabbitai[bot]`, `gemini-code-assist[bot]`): that commit is AI-written.

7. Agent files in the repo · medium · agent was set up

- **Rules / guidance**: `CLAUDE.md`, `AGENTS.md`, `GEMINI.md`, `.cursorrules`, `.github/copilot-instructions.md` ... (see Part 2).
- **Config / settings**: `.claude/settings.json`, `.gemini/settings.json`, `.aider.conf.yml`, `opencode.json` ...
- **Sub-agents, commands, skills, prompts**: `.claude/agents/`, `.claude/commands/`, `.claude/skills/`, `.github/agents/`, `.github/prompts/*.prompt.md`, `.gemini/commands/`, `.cursor/commands/` — a sign of heavier use.
- The date the file was **first added** (`git log --diff-filter=A -- CLAUDE.md`) estimates when adoption started.
- Files show the agent was set up, not who wrote a given commit. Over 40% of repos with agent files have no agent commit.

8. MCP configuration · medium · agent tooling

- Model Context Protocol servers give agents extra tools: `.mcp.json` (~79K repos), `.cursor/mcp.json`, `.vscode/mcp.json`, `.gemini/settings.json` (mcpServers).

9. CI workflows that run agents · strong · agent automated in the repo

- `anthropics/claude-code-action` (~20K workflow files), `openai/codex-action` (~1.3K), `google-github-actions/run-gemini-cli` (~1.2K), the `cursor-agent` CLI (~500).
- `.github/workflows/copilot-setup-steps.yml` (~6.8K) prepares the Copilot coding agent's environment.
- Workflow files named `claude*.yml`, `codex*.yml`, `gemini*.yml`.

10. Dependencies and dev environment · weak to medium

- npm packages: `@anthropic-ai/claude-code`, `@openai/codex`, `@google/gemini-cli`, `@augmentcode/auggie`.
- Dev container features: `ghcr.io/anthropics/devcontainer-features/claude-code`; Dockerfiles that install an agent CLI.
- Careful: a repo can depend on these because it **builds on** the agent (a plugin), not because it uses it.

11. Hidden use: ignore files and disabled traces · strong sign, but no commit-level data

- `.gitignore` entries for agent files (`CLAUDE.md`, `.cursor/`, `.aider*`, `.claude/settings.local.json`). ~20% of repos with agent files hide them this way. GitHub's own gitignore template `Global/Agents.gitignore` exists, so a copied template is weaker evidence.
- Settings that switch traces off: `"includeCoAuthoredBy": false` or an empty `attribution` in `.claude/settings.json`; aider `--no-attribute-author`; guidance files saying “never add Co-authored-by”. These repos use agents but leave no commit traces.

12. Behaviour without any marker · weak · estimate only

- Edit velocity (big changes in very little time), bursts of commits, long structured commit messages, emoji, conventional commits, code and tests changed together, a jump in activity after the first agent file.
- Use these only as **classifier features** trained on commits that do have markers, and validate by hand. Never as proof.

13. Outside the repository · not mineable at scale

- Dotfiles repos with a developer's personal agent config, surveys, and vendor telemetry. Useful context, not MSR data.

> **Adoption date of a repo** = the earliest of: first agent commit, first agent PR, first agent file added, first agent workflow added. Report which signal gave the date.

Part 2 — Every agent and its traces

Grouped by kind. The five agents in your methodology come first. A missing line means no reliable trace of that kind was found.

The five methodology agents

Claude Code · Anthropic

- **Commit trailer:** `Co-Authored-By: Claude <noreply@anthropic.com>` (~8.9M). The name is often the model: `Claude Opus 4.8`, `Claude Sonnet 4.6`.
- **Other trailers:** `Claude-Session: https://claude.ai/code/session_<id>` (Claude Code on the web); `Assisted-by: Claude Code:<model>`.
- **Commit author:** `Claude <noreply@anthropic.com>` (~21.9M).
- **Footer (commits and PR bodies):** `(robot emoji) Generated with [Claude Code](https://claude.com/claude-code)` (older: `claude.ai/code`). ~3.6M commits, ~19.5M PRs.
- **Bot:** `claude[bot]` (id 209825114) — the Claude GitHub Action. Opens PRs (~68K), comments on issues, co-authors commits (~96K).
- **Branch:** `claude/<slug>-<session id>` (Claude Code on the web).
- **Files:** `CLAUDE.md` (any folder), `.claude/` (settings.json, agents/, commands/, skills/), `.claudeignore`, `.mcp.json`.
- **CI / other:** `anthropics/claude-code-action` workflows; devcontainer feature `devcontainer-features/claude-code`; npm `@anthropic-ai/claude-code`; `@claude` mentions.
- **Hidden use:** `.claude/settings.json` with `includeCoAuthoredBy: false` or empty `attribution`.
- **Watch out:** other tools reuse Claude Code's trailer with their own model name, e.g. `Kimi Code <noreply@anthropic.com>`. The email says the tool is Claude Code.

GitHub Copilot · GitHub / Microsoft

- **Coding agent commits:** author `Copilot <198982749+Copilot@users.noreply.github.com>` (~4.4M) or `copilot-swe-agent[bot]`; the human is the co-author.
- **Trailer:** `Agent-Logs-Url: https://github.com/<owner>/<repo>/sessions/<id>` (since March 2026, ~724K).
- **PRs:** author `Copilot` (~2.1M PRs), branch `copilot/<slug>` or `copilot/fix-<ids>`. Started by assigning an issue to Copilot or `@copilot` in a comment.
- **VS Code co-author:** `Co-authored-by: Copilot <223556219+Copilot@users.noreply.github.com>` (setting git.addAICoAuthor). Weaker: added by mistake to human commits in VS Code 1.117–1.118 (late Apr–early May 2026).
- **Review:** `copilot-pull-request-reviewer[bot]` (~6.3M PRs reviewed) — review adoption.
- **Files:** `.github/copilot-instructions.md`, `.github/instructions/*.instructions.md`, `.github/prompts/*.prompt.md`, `.github/agents/`, `.copilotignore`.
- **CI:** `.github/workflows/copilot-setup-steps.yml`.
- **Old:** `copilot-workspace.githubnext.com` links (Copilot Workspace, 2024–25).

OpenAI Codex · OpenAI

- **Commit trailer:** `Co-authored-by: Codex <codex@openai.com>` (~274K). **Author:** any name with `codex@openai.com` (~640K).
- **PRs:** branch `codex/<slug>` (prefix can be changed), label `codex` (~4.5M), body link `chatgpt.com/codex/tasks/task_e_<id>` (~2M PRs), e.g. “Built by a Codex Cloud task”.
- **Bots:** `chatgpt-codex-connector[bot]` (199175422, reviews and PRs), `openai-codex[bot]` (215057067).
- **Files:** `.codex/`; `AGENTS.md` started with Codex but is now generic.
- **CI / other:** `openai/codex-action` workflows; npm `@openai/codex`; `@codex` mentions (e.g. `@codex review`).

Cursor · Anysphere

- **Cloud agent commits:** author `Cursor Agent <cursoragent@cursor.com>` (~2.3M), the human as co-author.
- **Trailers:** `Co-authored-by: Cursor <cursoragent@cursor.com>` (~3.4M); `Made-with: Cursor` from the IDE/CLI agent (~1.5M).
- **PRs:** author `cursor[bot]` (~176K), branch `cursor/<slug>-<4 hex>`, body `Made with [Cursor](https://cursor.com)` and `cursor.com/agents` links (~1.6M).
- **Review:** Bugbot reviews also come from `cursor[bot]`.
- **Files:** `.cursor/` (rules/*.mdc ~196K files, commands/, mcp.json), `.cursorrules` (older), `CURSOR.md`, `.cursorignore`.
- **CI / other:** `cursor-agent` CLI in workflows; `@cursor` mentions.

Gemini · Google

- **Gemini Code Assist:** bot `gemini-code-assist[bot]` (176961590). Mostly reviews (~1.7M PRs); co-author on commits that apply its suggestions (~374K).
- **Gemini CLI footer:** `(robot emoji) Generated with [Gemini CLI](https://github.com/google-gemini/gemini-cli)`; older co-author names `Gemini 2.5 Pro`, `Gemini 2.5 Flash`; `AI-Assisted-By: Gemini ...`.
- **Files:** `GEMINI.md`, `.gemini/` (settings.json, commands/).
- **CI / other:** `google-github-actions/run-gemini-cli` workflows; npm `@google/gemini-cli`; `@gemini-cli` mentions.
- **Related Google agents** (listed below): Jules, Antigravity, Stitch.

Cloud and autonomous agents (work through PRs)

Jules · Google

- **Bot:** `google-labs-jules[bot]` (161369871) authors commits (~1.3M) and PRs (~334K); also as co-author.
- **Branch:** no fixed prefix in 2026: `jules-<number>-<hash>` or `<slug>-<long number>`. Match the bot instead.
- **Files:** `.jules/`. **Other:** `@jules` mentions; rare `jules.google.com` links.

Google Antigravity · Google

- **Trailer:** `Co-authored-by: Antigravity <noreply@google.com>` (~32K); user `antigravity-ai` (256725992).
- **Footer:** `(robot emoji) Generated with [Antigravity CLI](https://antigravity.google)`.
- **Files:** reads `GEMINI.md` / `AGENTS.md` (no own file found).

Devin · Cognition

- **Bot:** `devin-ai-integration[bot]` (158243242): commits (~469K), PRs (~313K), co-author (~800K).
- **Branch:** `devin/<unix time>-<slug>`. **PR body:** `app.devin.ai/sessions/<id>` ("Link to Devin run").
- **Files:** `.devin/`. **Other:** `@devin` mentions.

OpenHands · All Hands AI

- **Author / trailer:** `openhands <openhands@all-hands.dev>` (~261K). **User:** `openhands-agent` (not a [bot]).
- **Branch:** `openhands/`. **Files:** `.openhands/`, `.openhands_instructions`. Often started by assigning an issue.

Factory (Droid) · Factory

- **Bot:** `factory-droid[bot]` (138933559) as author and co-author (~132K).
- **Files:** `.factory/` (droids/, commands/).

Amazon Q Developer · AWS

- **Bot:** `amazon-q-developer[bot]` (208079219). **Branch:** `Q-DEV-issue-<n>-<time>`.
- **Files:** `.amazonq/`, `AmazonQ.md`.

Kiro · AWS

- **Bot:** `kiro-agent[bot]` (245459735), PRs (~30K). **Trailer:** `Co-authored-by: Kiro Agent <244629292+kiro-agent@users.noreply.github.com>`; `Generated-by: Kiro IDE`.
- **Files:** `.kiro/` (steering/, specs/) — ~42K repos.

Codegen

- **Bot:** `codegen-sh[bot]` (131295404). **Branch:** `codegen-bot/<slug>`.

Sweep

- **Bot:** `sweep-ai[bot]` (128439645). **Branch:** `sweep/<slug>`.

Charlie · Charlie Labs

- **Bot:** `charliecreates[bot]` (198680274); co-author `CharlieHelps <charlie@charlielabs.ai>`.
- **Files:** `.charlie/instructions/`.

Grok · xAI

- **Bot:** `grok-by-xai[bot]` (261865991), co-author name like `Grok 4.7`.

Sentry Seer · Sentry

- **Bot:** `seer-by-sentry[bot]` / `sentry-autofix[bot]` (157164994) — automatic bug fixes.

Tembo

- **Bot:** `tembo[bot]` (208362400).

Gru

- **Author:** `gru-agent`. **Branch:** `gru/`.

Open SWE · LangChain

- **Author / trailer:** `open-swe`. **Branch:** `open-swe/`. **Label:** `open-swe`.

Abacus.AI

- **Bot:** `abacus-ai-automation[bot]` (194430910).

Microsoft Amplifier · Microsoft

- **Trailer:** `Co-authored-by: Amplifier <240397093+microsoft-amplifier@users.noreply.github.com>`.

Terminal and IDE agents (work in your repo, sign commits)

Aider

- **Author name:** `(aider)` added to author and committer name by default, e.g. `Jane Doe (aider)`.
- **Trailer (optional):** `Co-authored-by: aider (<model>) <aider@aider.chat>` — includes the model; older `aider <noreply@aider.chat>`.
- **Prefix (optional):** subject starts with `aider: `.
- **Files:** `.aider.conf.yml`, `.aiderignore`; aider adds `.aider*` to `.gitignore`. Do **not** use `CONVENTIONS.md` (too common).

Amp · Sourcegraph

- **Trailers:** `Co-authored-by: Amp <amp@ampcode.com>` (~124K) and `Amp-Thread-ID: https://ampcode.com/threads/T-<id>` (~138K).
- **Label:** `amp`. **Files:** `AGENT.md` (singular), `.sourcegraph/`.

OpenCode

- **Trailer:** `Co-authored-by: <model or OpenCode Agent> <noreply@opencode.ai>` (~93K). **Footer:** `(robot emoji) Generated with [OpenCode](https://opencode.ai)`.
- **Branch:** `opencode/`. **Files:** `.opencode/`, `opencode.json`.

Crush · Charm

- **Trailer:** `Assisted-by: <model> via Crush <crush@charm.land>`. **Footer:** `(heart emoji) Generated with Crush`.
- **Files:** `CRUSH.md`, `.crush/`.

Qwen Code · Alibaba

- **Trailer:** `Co-authored-by: Qwen-Coder <qwen-coder@alibabacloud.com>` (~116K). **Footer:** `(robot emoji) Generated with [Qwen Code](...)`.
- **Files:** `QWEN.md`.

Warp

- **Trailer:** `Co-Authored-By: Warp <agent@warp.dev>`. **Files:** `WARP.md`, `.warp/`.

Goose · Block

- **Trailer:** `Co-authored-by: Goose <opensource@block.xyz>`. **Files:** `.goosehints`. **Branch:** `goose/`.

Junie · JetBrains

- **Trailer:** `Co-authored-by: Junie <junie@jetbrains.com>`. **Bot:** `jetbrains-junie[bot]` (201638009).
- **Branch:** `jetbrains_junie…`. **Files:** `.junie/` (guidelines.md).

Cline

- **Trailer:** `Co-Authored-By: Cline <noreply@cline.bot>` (rare; off by default). **Footer:** `(robot emoji) Generated with [Cline](https://cline.bot)`.
- **Cloud:** bot `cline-cloud[bot]` (276134852), branch `cline/<id>`. **Files:** `.clinerules`, `.cline/`, `memory-bank/`.

Roo Code

- **Cloud bot:** `roomote-roomote[bot]` (301996811); older author `roomote`, `roomote@roocode.com`.
- **Files:** `.roo/`, `.roorules`.

Kilo Code

- **Name:** `Kilo Code` (users often type their own email, so match carefully). **Files:** `.kilocode/`, `.kilocodeignore`.

Windsurf (Cascade) · Cognition

- **Bot:** `windsurf-bot[bot]` (189301087). **Footer:** `(robot emoji) Generated with [Windsurf Cascade](...)` (rare).
- **Files:** `.windsurf/`, `.windsurfrules`, `.codeium/`.

Continue

- **Trailer:** `Co-Authored-By: Continue <noreply@continue.dev>`. **Files:** `.continue/`.

Augment Code (Auggie)

- **Trailer:** `Co-authored-by: Augment <noreply@augmentcode.com>`. **Files:** `.augment/`, `.augmentignore`, `.augment-guidelines`. **npm:** `@augmentcode/auggie`.

TRAE · ByteDance

- **Trailer:** `Co-authored-by: TRAE CLI <traecli@bytedance.com>`. **Files:** `.trae/`, `.traerules`, `.marscode/`.

Letta Code · Letta

- **Trailer:** `Co-Authored-By: Letta Code <noreply@letta.com>`.

Mistral Vibe · Mistral

- **Trailer:** `Co-Authored-By: Mistral Vibe <vibe@mistral.ai>`.

Ona · formerly Gitpod

- **Trailer:** `Co-authored-by: Ona <no-reply@ona.com>` (~52K).

Pi

- **Trailer:** `Co-authored-by: Pi <noreply@pi.dev>`. **Files:** `.pi/`.

Kimi Code · Moonshot

- **Trailer:** `Kimi Code <noreply@anthropic.com>` (Claude-Code-compatible harness). **Files:** `.kimi/`.

Sketch

- **Trailer:** `Co-authored-by: sketch <hello@sketch.dev>`.

CodeBuddy · Tencent

- **Trailer:** `Co-authored-by: codebuddy <codebuddy@users.noreply.github.com>`. **Files:** `CODEBUDDY.md`.

Blackbox AI

- **Trailer:** `Co-authored-by: blackboxai <blackboxai@users.noreply.github.com>`.

Tabnine, Zencoder

- **Trailers (rare):** `Tabnine CLI <noreply@tabnine.com>`, `Zencoder <noreply@zencoder.ai>`.

ChatGPT · OpenAI

- **Co-author name:** `ChatGPT` (Robbes et al.). Weak: people paste ChatGPT code without a standard trailer.

File-only agents · no commit trace found

- Alibaba Lingma `.lingma/` · Baidu Comate `.comate/` · Brokk `.brokk/` · Plandex `.plandex/` · Atlassian Rovo Dev `.rovodev/` · Qodo `.qodo/` · Verdent `.verdent/` · Tessl `.tessl/` (bot `tessl-helper[bot]`) · Serena `.serena/` · Rulesync `.rulesync/`.
- Zed's agent reads other agents' files and leaves no trace of its own.

App builders (“vibe coding” platforms)

Lovable · formerly GPT Engineer

- **Bot author:** `gpt-engineer-app[bot]` / `lovable-dev[bot]` (159125892) — ~39M commits.
- **Trailer:** `X-Lovable-Edit-ID: edt-<id>`; the human is the co-author.

Replit Agent · Replit

- **Trailers:** `Replit-Commit-Author: Agent` (~9.9M) and `Replit-Commit-Session-Id: <id>`.
- **Co-author:** `Replit Agent <agent@replit.com>`. **Files:** `replit.md`, `.replit` (the platform).

v0 · Vercel

- **Bot:** `v0[bot]`. **Trailer:** `Co-authored-by: v0 <it+v0agent@vercel.com>`.

Google Stitch · Google

- **Bot:** `google-labs-stitch[bot]` (315546542) — UI design agent.

Bolt.new · StackBlitz

- No reliable commit trace found; projects may contain a `.bolt/` folder.

Code-review and fix bots (review adoption, not authorship)

Reviewers

- `coderabbitai[bot]` (136622811) · `copilot-pull-request-reviewer[bot]` · `gemini-code-assist[bot]` · `cursor[bot]` (Bugbot) · `chatgpt-codex-connector[bot]` · `qodo-code-review[bot]` (151058649) / `qodo-merge-pro[bot]` · `ellipsis-dev[bot]` (65095814) · `greptile-apps[bot]` · `sourcery-ai[bot]` · `deepsource-autofix[bot]`.
- A commit with one of these as **co-author** applied an AI fix (e.g. `coderabbitai[bot]` ~129K): count it as AI-written.

Generic traces (some agent, unknown which)

Any agent

- **Files:** `AGENTS.md` (~1.2M files; read by Codex, Copilot, Cursor, Jules, Aider, Gemini and more), `AGENT.md`, `llms.txt` (weak).
- **Trailers:** `Assisted-by:`, `AI-Assisted-By:`, `Generated-by:` (check the tool named).
- **Label:** `ai-generated` (~30K PRs).
- **Planning tools used with agents:** SpecKit `.specify/` · Taskmaster `.taskmaster/` · MCP `.mcp.json`.

Part 3 — What not to count, and traps

- **Automation bots are not agents:** `dependabot[bot]`, `renovate[bot]`, `github-actions[bot]`, `snyk-bot`, `pre-commit-ci[bot]`, `codecov[bot]`, `mergify[bot]`, `release-please[bot]`, `semantic-release-bot`. Remove them before counting human vs AI commits.
- **Brand words are not signals:** “move the cursor” is not Cursor; a person called Claude or Jules is not an agent; the login `osman-haider` is not Aider.
- **Trust the email, not the name:** `Kimi Code <noreply@anthropic.com>` is Claude Code; `Claude Opus 4.8 <noreply@anthropic.com>` is still Claude Code.
- **Made-up identities:** some people type `kilo@example.com` or `roo-code@example.com`. Count only the official addresses above.
- **Spoofed bots:** we saw `claude[bot]` commits carrying github-actions' id (41898282). Check the id.
- **VS Code Copilot bug:** `Co-authored-by: Copilot` (223556219) was added to human commits in VS Code 1.117–1.118 (late Apr–early May 2026).
- **Generic files:** `AGENTS.md` shows that some agent is used, not which one. `CONVENTIONS.md` and `llms.txt` are mostly not agent files.
- **Squash merges:** keep only the last author, so a PR's agent label must be copied to the squash commit.
- **Outsider PRs:** an agent PR from a stranger (even spam) is not adoption by the project. Count merged agent PRs separately.
- **Building on vs using:** a repo can contain agent packages or workflows because it is a plugin or a fork of the agent.
- **How accurate?** Robbes et al. hand-checked ~1,200 artifacts: 0.5–1% false positives, ~3% borderline. Misses are far more common than false alarms.

Part 4 — Edit velocity

> **edit velocity = (lines added + lines deleted) ÷ seconds since the same author's previous commit**

- **Get the data:** `git log --numstat --format=%H|%ae|%aI|%P`. Add up added + deleted lines over all files (binary files show `-`: count 0).
- **Group** commits by author email (lowercase) and sort each author's commits by **author date** (not committer date, which rebases and squashes reset).
- **Gap** = this commit's author date minus the same author's previous one, in seconds. An author's first commit has no value; skip gaps of 0 or less.
- **Leave out of the ranking:** merge commits (2+ parents, no line counts), squash merges (title ends with `(#N)`), lockfile-only commits, automation bots.
- **Flag** the fastest 5% (95th percentile). Decide once whether the percentile is per repo or across the whole corpus.
- **Example:** +600 lines, 90 s after the previous commit = 6.7 lines per second, far faster than typing → flagged. A 2,000-line `package-lock.json` change is faster still, but left out.
- **Pitfalls:** tiny gaps (1 s) give huge values — use a minimum gap such as 60 s, or log(velocity). Agent-authored commits (author = Copilot, Cursor Agent ...) form their own author group. Remove trailers and footers before computing message features.
- Edit velocity is a **feature for the classifier**, not proof of an agent.

Sources

- Robbes, Matricon, Degueule, Hora, Zacchiroli. “Agentic Much? Adoption of Coding Agents on GitHub.” ACM TOSEM 2026 (arXiv 2601.18341); “Promises, Perils, and (Timely) Heuristics for Mining Coding Agent Activity.” MSR 2026 (arXiv 2601.18345). Heuristics: github.com/labri-progress/agent-impact.
- Our GitHub commit, PR, issue and code searches on 24–25 Oct 2026 for every trace listed (counts are GitHub's approximate totals).
- Aider docs (git attribution); GitHub changelog 20 Mar 2026 (Agent-Logs-Url); Cursor forum staff answers (Cursor Agent, cursor/ branches, Made-with); Linux kernel docs, AI Coding Assistants (Assisted-by); VS Code issue 314311 and press (git.addAICoAuthor, Apr–May 2026).

Traces change every few months. Re-run the searches before a new study.